

PX-057

Adaptive Stopping to Prevent LLM Overthinking

Executive determination, evidence summary, and investment recommendation

FINAL PRAXIS REPORT / DECISION BRIEF

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Experiment stage: Gate 2 complete; H1-H3 passed; H4 transfer pending

CLAIM BOUNDARY

This document reports one frozen 200-item GSM8K sample, Qwen2.5-7B-Instruct, and one eight-round iterative prompting protocol. It does not claim cross-model, cross-domain, or large-scale robustness.

Final Determination

GO FOR H4 REPLICATION. PX-057 is a valid positive Gate 2 result and the strongest current candidate for further investment. All preregistered H1-H3 checks passed. Do not represent the result as large-scale or general robustness until cross-model and cross-domain transfer is demonstrated.

Preregistered Gate 2 outcomes

Accuracy delta	+29.5 pp	threshold: >= -1.0 pp	PASS
Compute saving	66.5%	threshold: >= 20%	PASS
Prevention rate	89.6%	threshold: >= 25%	PASS
Early-stop harm	0.5%	threshold: <= 2%	PASS

All four registered checks passed; H4 transfer was outside this gate.

Registered outcomes and decision thresholds.

KPI	Result	Decision meaning
Adaptive accuracy	91.0% (182/200)	29.5 pp above fixed-long
Fixed-long accuracy	61.5% (123/200)	Eight reconsideration rounds degraded many answers
Fixed-short accuracy	88.0% (176/200)	Adaptive added 3.0 pp over round 2
Compute saving	66.5% [63.7%, 69.1%]	Passed 20% floor
Prevention	60/67; 89.6% [82.1%, 95.5%]	Passed 25% floor
Early-stop harm	1/200; 0.5%	Passed 2% ceiling

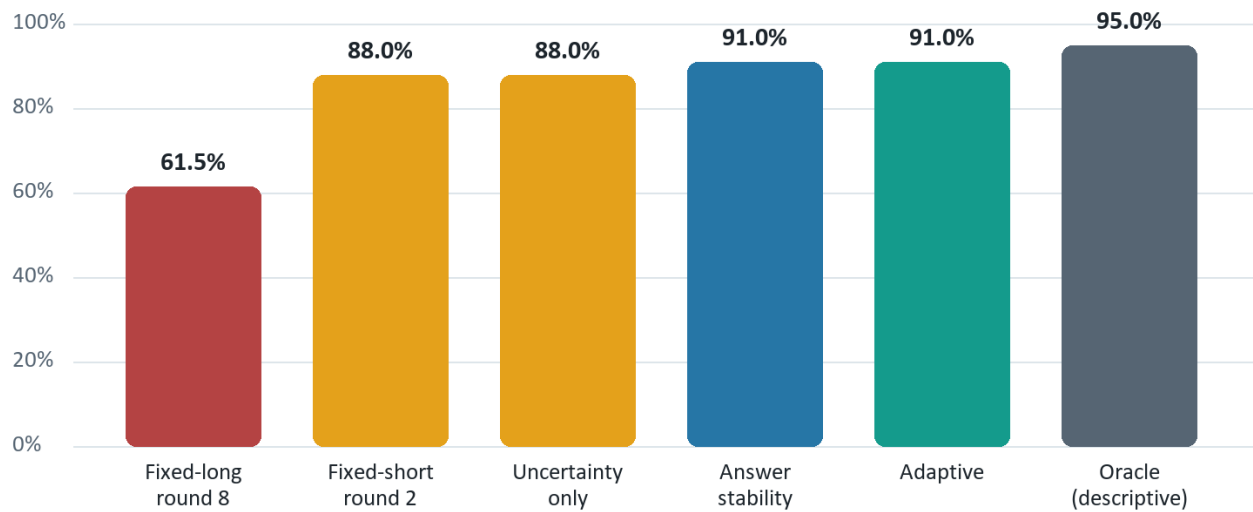
What the Result Proves

- The frozen Qwen/GSM8K iterative protocol exhibited frequent correct-to-wrong degradation by round 8.
- The preregistered adaptive policy avoided most observed degradation and used substantially fewer generated tokens.
- The evidence package is complete: 200 unique questions, 200 complete traces, and 1,600 unique round generations were independently verified.

Critical Nuance

Adaptive stopping and answer-stability stopping both scored 91.0%. The final Praxis should therefore be framed around stability-gated stopping, not around a proven benefit from the confidence threshold. The confidence component remains a hypothesis for H4, not an established mechanism.

Accuracy by inference policy



The adaptive arm tied the simpler stability-only control.

What It Does Not Prove

- No full-GSM8K or population-level accuracy estimate.
- No cross-model, cross-domain, stochastic-decoding, or prompt-transfer result.
- No calibrated uncertainty result.
- No direct latency, energy, or dollar-cost result.

Investment Recommendation

Invest in one tightly frozen H4 study before expanding implementation. Preserve the current result as the discovery experiment and preregister the transfer matrix before inspecting new outputs.

1. Replicate on a second open model without retuning the current policy on the test set.
2. Add a non-math corpus with deterministic or blinded scoring.
3. Keep all six comparison arms, especially stability-only.
4. Add latency, GPU-seconds, and dollar cost.
5. Retain confidence only if it improves accuracy, reduces harm, or lowers cost versus stability-only.

Publication-Ready Claim

On a frozen 200-item GSM8K sample with Qwen2.5-7B-Instruct and an eight-round iterative reconsideration protocol, a preregistered adaptive stopping rule achieved 91.0% accuracy versus 61.5% for fixed-long inference, saved 66.5% of generated tokens, prevented 60 of 67 observed correct-to-wrong events, and harmed 1 of 200 questions. These findings are limited to the tested model, sample, prompting protocol, and thresholds.

Literature Foundation

Zhou, S., et al. (2026). When More Thinking Hurts: Overthinking in LLM Test-Time Compute Scaling. Findings of ACL 2026. [Read the original publication](#)

Cobbe, K., et al. (2021). Training Verifiers to Solve Math Word Problems. Introduces GSM8K. [Read the dataset paper](#)